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Potential as “intrinsic energy of inertia”

Electro-Magnetic Field (force caused by potential)

- static: localized source charge at rest, only spacial dependency, no time dependency

- stationary: source charge at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistic: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movement
- vacuum vs. matter
- local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := \text{rot}(A) \stackrel{\text{Matter}}{=} \mu_0 (\mathcal{H} + \mathcal{M}) \stackrel{\text{Vacuum}}{=} \frac{1}{c} \left(\frac{r}{|r|} \times \mathcal{E} \right) \stackrel{\text{Far Field}}{=} \frac{1}{2} \frac{1}{2\pi} \mu_0 \frac{k^2 e^{i(kr - \omega t)}}{r} \frac{r}{|r|} \times \left(c p_{0\text{dipol}} + m_0 \right) \quad (1)$$

Magnetic Field in Matter

Excitation $\hat{=}$ Displacement

$$\text{Excitation:} \quad \mathcal{H} = \frac{1}{\mu_0} \mathcal{B} - \mathcal{M} = \frac{1}{\mu} \mathcal{B} \quad \text{Magnetisation:} \quad \mathcal{M} = \frac{\mu - \mu_0}{\mu \mu_0} \mathcal{B} = \frac{\mu - \mu_0}{\mu_0} \mathcal{H} \quad (2)$$

Electric Field in Vacuum

$$\mathcal{E} \stackrel{\text{conservative}}{=} -\nabla U_e \stackrel{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \stackrel{\text{fast moving charge}}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\varepsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|} \cdot \frac{v}{c}\right)\right]}{r}}_{\text{acceleration}} \right] \quad (3)$$

Electric Field in Matter

Displacement $\hat{=}$ Excitation (field change)

dielectricum = non-conducting material

electricum = conductor medium

$$\text{Excitation: } \mathcal{D} = \varepsilon_0 \mathcal{E} + \mathcal{P} = \overset{\text{isotropic medium in weak fields}}{=} \varepsilon_0 \mathcal{E} + \varepsilon_0 \chi \mathcal{E} = \varepsilon_0 (1 + \chi) \mathcal{E} = \underset{(4)}{\varepsilon_0 4\pi r^3 n \langle p_m \rangle = 4\pi r^3 n}$$

Abbreviations

Φ = electric potential A = magnetic potential $j = \nabla \times \mathcal{H}$ = elec. current density

$\mathcal{E}$ = Electric Field $\mathcal{B}$ = Magnetic Field $\nabla \cdot \mathcal{B} = 0$ $\mathcal{D}$ = Displacement density of dielectricum $\mathcal{M} =$

$\mathcal{P}$ = Polarisation $\mathcal{H}$ = magn. excitation, magn. field strength ("magn. displacement")

n = spacial molecule density in volume

$\alpha = 4\pi r^3 = \text{constant}$ χ = elec. susceptibility of dielectricum $\kappa = \frac{\mu - \mu_0}{\mu_0} = \text{magn. susceptibility}$

$\langle p_m \rangle$ = spacial average of magn. dipol moment $\langle \mathcal{E}_m \rangle$ = spacial average of electric field

$\bar{\mathcal{E}}_m$ = time average of electric field μ = magn. permeability μ_0 = magn. constant ...

ε = dielectricity constant $\varepsilon_0 = \dots$

Electro-Magnetic interactions/coupling (Maxwell conditions)

conservativ, static charge localisation:

...

stationary, slow charge movement:

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \quad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t} \mathcal{B} = i\omega \mathcal{B} \quad \nabla \times \frac{1}{\mu_0} \mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t} \mathcal{E} = j - i\omega \varepsilon_0 \mathcal{E} \quad (5)$$

relativistic, fast charge movement:

...

Potential

⇒ Wave equation that determines Potential:

$$\Delta \cdot U - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} U = J \quad (6)$$

⇒ Potential caused by “ignition J” (inhomogeneity) from source at rest, $v = 0$:

$$U = -\frac{1}{4\pi} \int \frac{J}{|r|} dr' \quad (7)$$

elec. Potential structure:

$$J = -\frac{1}{\varepsilon_0} \varrho$$

$$U_e := \Phi \stackrel{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi} \frac{1}{\varepsilon_0} \int \frac{\varrho}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} Q = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} It \quad (8)$$

magn. Potential structure:

$$J = -\mu_0 j$$

$$U_m := A \stackrel{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi} \int \frac{j}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \mu_0 I = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{c^2 \varepsilon_0} I = \frac{1}{c^2} \frac{1}{t} U_e \stackrel{\text{source in uniformal movement, } v=\text{const}}{=} \quad (9)$$

Electric endogen energy

- Bond Energy
- Binding Energy

$$E_{pot} = mc^2$$

$$e = \text{electron charge} \quad q = ne = \text{charge} \quad n \in \mathbb{N} \quad (10)$$

$$\text{Potential} \quad U = \frac{E_{pot}}{q} = \frac{(mc^2)^2}{ne} = \frac{m}{q} mc^4 = \kappa mc^4 \quad (11)$$

Electric induced energy (exogenous excitation)

- Ionisation Energy
- Excitation Energy

$$E_{kin} = \frac{1}{2} mv^2 \hat{=} (pc)^2 \hat{=} h\nu$$

E_{kin} Energy density electric:

$$w_e = \frac{1}{2} \varepsilon_0 \mathcal{E}^2 \quad (12)$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \quad (13)$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2} CU^2 \quad (14)$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2} LI^2 \quad (15)$$

Power:

$$P = VI = (U_1 - U_2) \frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \quad (16)$$

Frequency:

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda \frac{1}{T} \stackrel{vacuum}{=} kc \stackrel{matter}{=} k \frac{c}{\eta} = kc \frac{\tan \beta}{\tan \alpha} \hat{=} kc \left(\mathcal{E}_{\beta}^{-1} \mathcal{E}_{\alpha} \right)^{-1} = kc \frac{\varepsilon_{\beta}}{\varepsilon_{\alpha}} = kc \sqrt{\frac{\varepsilon_0 \mu_0}{\varepsilon \mu}} \hat{=} \sqrt{\frac{k}{m}} \hat{=} \frac{1}{\sqrt{LC}} \quad (17)$$

Phase:

$$\varphi = k \left(\vec{r} \cdot \vec{k} \right) - \omega t := ks - \omega t = \dots \quad (18)$$

Refraction:

Refraction $\Leftrightarrow \exists$ Matter; refraction causes dispersion

$$\eta = \frac{\tan \alpha}{\tan \beta} \hat{=} \mathcal{E}_{\beta}^{-1} \mathcal{E}_{\alpha} = \frac{\varepsilon_{\alpha}}{\varepsilon_{\beta}} = \dots \quad (19)$$

Dispersion:

Dispersion $\Leftrightarrow \exists$ Matter (dielectric medium or conductor); Dispersion $\neq$ Dissipation

$$\dots \quad (20)$$

Dielectricity:

$$\varepsilon = \dots \quad (21)$$

Permeability:

$$\mu = \dots \quad (22)$$

Conductivity:

$$\sigma = \dots \quad (23)$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \varepsilon_0 \quad (24)$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c} \frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \quad (25)$$

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t} (we + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) \quad (26)$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \quad (27)$$

$$\Delta \cdot := \text{div grad} = \text{Laplacian}$$

J = "ignition of inhomogeneity, cause" j = Current density ϱ = Charge density

V = Potentialdifference I = Current R = Resistance

L = Inductance C = Capacitance Q = charge (ion) $\Psi = LI$ = mag. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H}$ = Poynting-Vector = radiation intensity

$D_S = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr$ = dissipation energy lost from radiation ≈ 0 for low frequencies ω

$$d = \sqrt{\frac{1}{\mu\sigma\omega}} = \text{skin depth (surface) of conductor} \quad r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho = \text{radius (extension) of conductor}$$

$$\rho = kr = \text{"radius of curvature } k\text{"}$$

$$\mu = \text{permeability of conductor} \quad \sigma = \text{conductivity of conductor (dielectricum)}$$

Low Frequency

$$\omega \ll \frac{1}{\mu\sigma r^2} \iff r \ll d \iff \rho \ll 1 \quad (28)$$

$$L \approx \text{const.} \quad C \approx \text{const.} \quad D_S \approx 0$$

High Frequency

$$\omega \gg \frac{1}{\mu\sigma r^2} \iff r \gg d \iff \rho \gg 1 \quad (29)$$

$$L(t) \neq \text{const.} \quad C(t) \neq \text{const.} \quad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

Continuity at Rest: Electric Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emergence of Radiation

Continuity and periodic (harmoneous oscillation) Movement in VACUUM

$$2\pi = k\lambda \gg kr_0 \ll 1$$

$$\omega = kc \ll \frac{c}{r_0} \quad \nabla \cdot j = i\omega \varrho \quad i\omega Q = 0$$

Continuity and periodic (harmoneous oscillation) Movement in MATTER

Refraction index emerges

Dispersion: $k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$

Near Field

$$r \ll \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \quad kr \ll 2\pi$$

$$\frac{w_m}{w_e} \sim \frac{\frac{\vec{p}^2}{\varepsilon_0 r^6}}{\frac{k^2 \mu_0 c^2 \vec{p}^2}{r^4}} \sim (kr)^2 \ll 1$$

$\vec{p}$ = elec. Dipol

Far Field

accelerated movement -> Radiation in Far Field of charge source

$$r \gg \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \quad kr \gg 1 \quad r \gg kr_0^2 \ll 1$$

$$\frac{w_m}{w_e} = \frac{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|} \right)^2}{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|} \right)^2} = 1$$

Dissipation in Vacuum

no Continuity of charge/matter, no Refraction index, Reflection

Dissipation in Matter

no Continuity of charge/matter with Dielectricum, Refraction merges, Reflection, Absorption

Energy gain by excitation:

Action Quantum, Planck Constant (Energy-Frequency-Slope)¹:

$$h = \tan(\alpha) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \quad (30)$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \quad (31)$$

$$E_{kin}^{max} = eV_0 = e \frac{Q}{C} = h(\nu - \nu_0) \quad (32)$$

V_0 = Opposing Potential ($I = 0$)

$\alpha = \angle(E_{kin}, \nu)$ e = electron charge Q = ion charge

$$C = \text{Capacitance} \quad \nu = \frac{1}{2\pi} \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{C}{L} R^2}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C} \left(\frac{e}{h} Q + \nu_0 \right)^2$$

¹Wirkungsquantum

$\Rightarrow$ Ionisation Energy of chemical element $E_i = h\nu = h\frac{\lambda}{c} = \hbar\omega = \frac{\hbar}{k\lambda}\omega$

$\Rightarrow$ Energy released (gained) $E_{gain} = E_{pot} - E_i = mc^2 - h\nu$